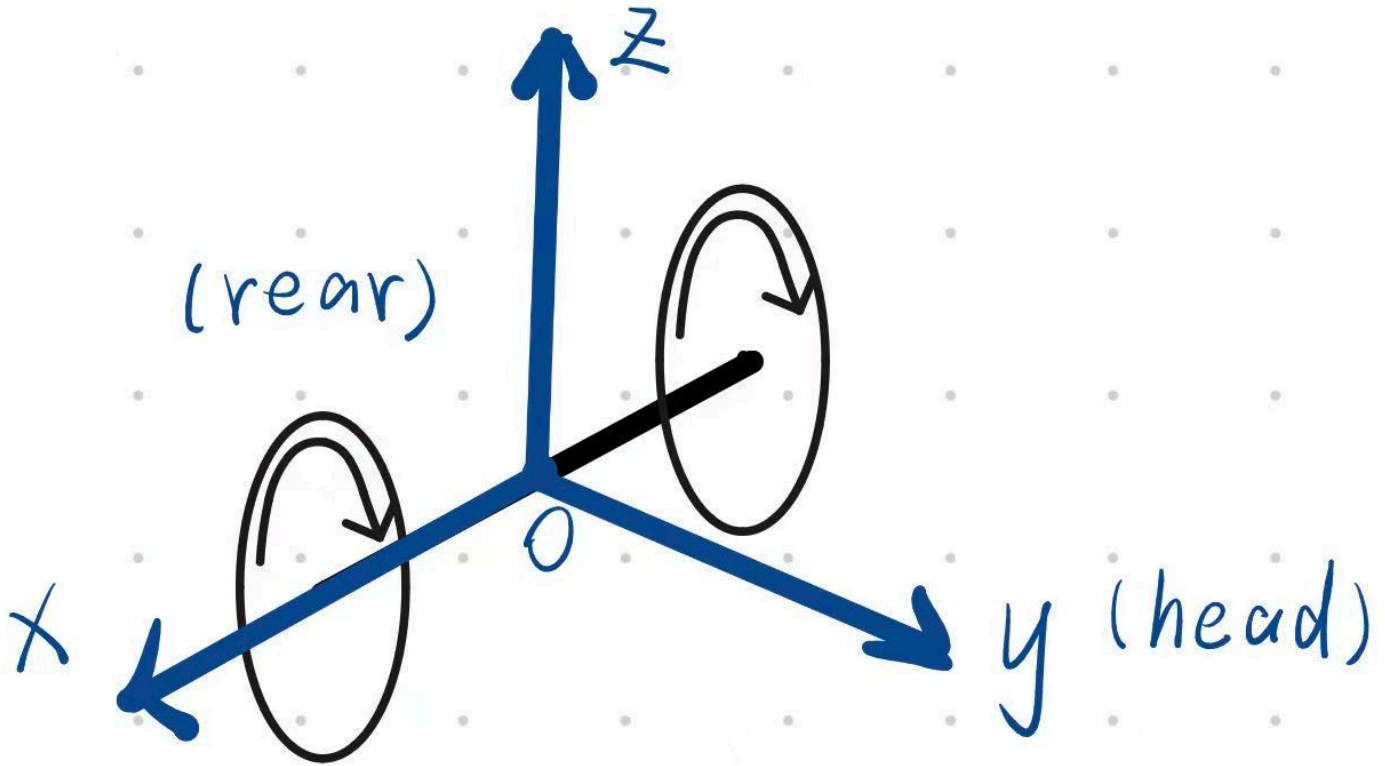


Attitude Estimation Algorithm

Coordinate System Definition

Due to the fixed installation of MPU6050, a body coordinate system is determined as follow:



$$\begin{cases} \dot{\phi} = p + (r \cos \phi + q \sin \phi) \tan \theta \\ \dot{\theta} = q \cos \phi - r \sin \phi \\ \dot{\psi} = \frac{1}{\cos \theta} (r \cos \phi + q \sin \phi) \end{cases}$$

p, q, r are three angular velocity orbiting around three axis of the body coordinate system and they could be measured by MPU6050 directly (designated as gx, gy, gz in code).

In this project, mechanical design of the car results in zero-pitch/ θ . Thus $\dot{\phi} = p$, and $\dot{\psi} = r \cos \phi + q \sin \phi$.

If ϕ is very small, the latter could also be simplified as $\dot{\psi} = r$. However, it is not as applicable as the former. A better way to measure yaw is to use the encoders attached to the wheel.

The emphasis of this project is on keeping roll/ ϕ around zero.

Roll-Measurement: Complementary Filter

For an accelerometer, the output $\vec{f} = \vec{a} - \vec{g}$. Only when $\vec{a} = 0$, can ϕ be calculated as $\phi = \arctan_2 \frac{ay}{az}$. Actually non-zero $\vec{a}$ is caused by vibration, sudden launch and stop, which get high-frequency noise included; For a gyroscope, the static bias and errors accumulated gradually in every linear approximation step both act as low-frequency noise. Therefore, by applying a low-pass filter to the accelerometer and a high-pass filter to the gyroscope, then adding them together, we can gain a ideal measurement for ϕ .

LPF & HPF (one order)

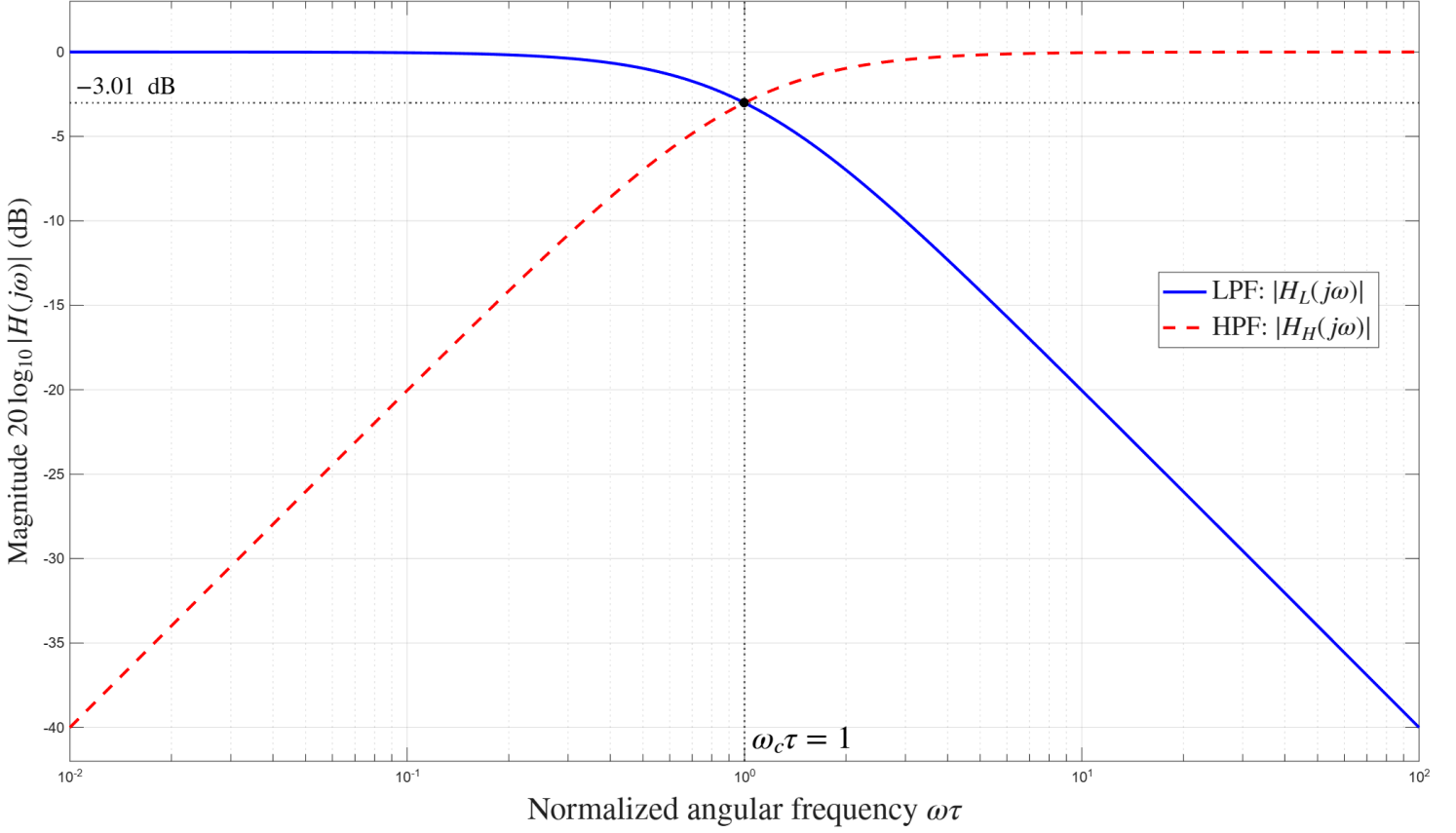
$$LPF : H_L(s) = \frac{1}{\tau s + 1}$$

$$HPF : H_H(s) = \frac{\tau s}{\tau s + 1}$$

$$H_L(s) + H_H(s) = 1$$

First-order LPF and HPF magnitude responses

$$\omega_c = 1/\tau \text{ rad/s, } f_c = 1/(2\pi\tau) \text{ Hz}$$



Theoretical Derivation

The basic idea is $\hat{\Phi}(s) = H_L(s)\Phi_a(s) + H_H(s)\Phi_g(s)$, where $\Phi_a(s)$ comes from the accelerometer, $\Phi_g(s)$ comes from the gyroscope and $\hat{\Phi}(s)$ is the estimated roll angle.

Expand the basic idea with expressions of LPF&HPF:

$$\hat{\Phi}(s) = \frac{1}{\tau s + 1}\Phi_a(s) + \frac{\tau s}{\tau s + 1}\Phi_g(s)$$

where $\Phi_g(s) = \frac{\Omega_g(s)}{s}$, due to $\phi = \int_0^t \omega_g(\tau)d\tau$. Furthermore,

$$\tau s \hat{\Phi}(s) + \hat{\Phi}(s) = \Phi_a(s) + \tau \Omega_g(s)$$

Then, we apply $\mathcal{L}^{-1}$ to the equation above, to get the time-domain equation:

$$\tau \frac{d\hat{\phi}}{dt} + \hat{\phi} = \phi_a + \tau \omega_g$$

Next, use backward difference $\frac{d\hat{\phi}}{dt} \approx \frac{\hat{\phi}[k] - \hat{\phi}[k-1]}{\Delta t}$ to replace derivative operation to change it into discrete form:

$$\hat{\theta}[k] = \frac{\tau}{\tau + \Delta t}(\hat{\theta}[k-1] + \omega_g[k]\Delta t) + \frac{\Delta t}{\tau + \Delta t}\theta_a[k]$$

(Sorted)

Define $\alpha = \frac{\tau}{\tau + \Delta t}$ to simplify this equation as:

$$\hat{\theta}[k] = \alpha \left(\hat{\theta}[k-1] + \omega_g[k]\Delta t \right) + (1 - \alpha)\theta_a[k]$$

In this project, sampling time $\Delta t = 0.005s$ and $\tau = 0.1s$ (which means ω_c is specified as 10 rad/s), so that $\alpha = 0.95238$.

Yaw-Measurement: Encoder Difference